

Curriculum Vitae

Arhit Chakrabarti

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Education

Texas A&M University, College Station, Texas, **Ph.D. in Statistics** 2021 – 2026

- **Advisors:** Dr. Bani K. Mallick and Dr. Yang Ni
- **Research Interest:** Bayesian Nonparametrics, Machine Learning, Spatial Transcriptomics, Unsupervised learning.
- GPA: 4.0/4.0

University of Calcutta, Kolkata, India, **M.Sc. in Statistics** 2015 – 2017

- **Project:** Statistical Inference with Standard Adaptive Designs and a Proposed Alternative
- GPA: 92.2/100 (First in First Class)

Presidency University, Kolkata, India, **B.Sc. in Statistics** 2012 – 2015

- **Minors:** Mathematics and Economics
- GPA: 9.5/10 (First in First Class)

Experience

Postdoctoral Fellow, [The University of Texas at Austin](#) – Department of Statistics and Data Sciences 2026 – Current

Graduate Teaching and Research Assistant, [Texas A&M University](#) 2021 – 2026

- Working as an RA for variety of projects mainly arising from the field of Bioinformatics.
- Worked as TA for STAT 600 (PhD level Statistical Computation course).
- Worked as TA for STAT 303 (Undergraduate Introductory Statistics course).
- Lecturer for STAT 201: Elementary Statistical Inference.

Senior Biostatistician, [Novartis Healthcare Pvt. Ltd.](#) – Early Development Biostatistics 2019 – 2021

- Worked extensively on developing Bayesian Hierarchical Regression models for dose-finding studies and supported dose escalation decisions, development of the study design, preparation of statistical analysis plan and performing exploratory analyses.
- Worked on the development of exploratory visualizations to better understand the data and have successfully provided answers to the Health Authority questions and requests.

Biostatistician, [Novartis Healthcare Pvt. Ltd.](#) – Global Medical Affairs 2017 – 2019

- Supported the statistical activities of clinical trials and adhoc exploratory analysis.
- Supported the statistical activities for publications in manuscripts and posters apart from preparation of statistical analysis plan and development of study designs.

Publications

Nested Atoms Model with Application to Clustering Big Population-Scale Single-Cell Data 2026

Chakrabarti, A., Ni, Y., Jiang, Y. Mallick, B

Accepted with minor revisions, *The Annals of Applied Statistics*

[\[Third Place Award at 2026 Best of Statistical Science \(BOSS\) Workshop\]](#)

LILRB4 regulates circadian disruption-induced mammary tumorigenesis via non-canonical WNT signaling pathway 2025

Ogunlusi, O., Sarkar, M., Carter, K., **Chakrabarti, A.**, Boland, D. J., Nguyen, T., et. al
Oncogene, 1-14, 2025.

Global-Local Dirichlet Processes for Clustering Grouped Data in the Presence of Group-Specific Idiosyncratic Variables 2025

Chakrabarti, A., Ni,Y., Pati, D., Mallick, B

Proceedings of the 42nd International Conference on Machine Learning, PMLR 267:7214-7249, 2025

[\[2025 SETCASA Poster Competition Golden Award\]](#)

Global-Local Dirichlet Processes for Identifying Pan-Cancer Subpopulations Using Both Shared and Cancer-Specific Data 2025

Chakrabarti, A., Ni,Y., Pati, D., Mallick, B

The Annals of Applied Statistics, 19(3), 2254-2278, September 2025

Joint Bayesian estimation of cell dependence and gene associations in spatially resolved transcriptomic data 2024

Chakrabarti, A., Ni,Y., Mallick, B

Scientific Reports, 14, 9516 (2024)

[\[Poster award, 6th Annual Symposium Cancer Research: Basic Science to Bioinformatics\]](#)

Graphical Dirichlet Process for Clustering Non-Exchangeable Grouped Data 2024

Chakrabarti, A., Ni,Y., Morris, E.R.A., Salinas, M.L., Chapkin, R.S., Mallick, B

Journal of Machine Learning Research, 24(323):1-56, 2024 [\[SBSS Student Paper Award 2024, American Statistical Association\]](#)

Initiation of sacubitril/valsartan shortly after hospitalisation for acutely decompensated heart failure in patients with newly diagnosed (de novo) heart failure: a subgroup analysis of the TRANSITION study 2020

Senni, M., Wachter, R., Witte, K. K., Straburzynska-Migaj, E., Belohlavek, J., Fonseca, C., Mueller, C., Lonn, E., **Chakrabarti, A.**, Bao, W., Noe, A., Schwende, H., Butylin, D., Pascual-Figal, D., and on behalf of the TRANSITION Investigators

European Journal of Heart Failure, vol. 22, no. 2, pp. 303-312, 2020.

Additional Academic Projects

Analysis of “Learn-As-You Go” (LAGO) study and extension for multi-stages and mixed intervention: Application to the ESSENCE study 2018 – 2019

- Worked as a Research Assistant, jointly under the collaboration of Department of Statistics, University of Calcutta, Department of Biostatistics, Harvard T.H. Chan School of Public Health, Department of Mathematics and Statistics, Boston University, Department of Statistics and Data Science, Yale University, Department of Statistics and Operations Research, Tel Aviv University.
- This was a large scale public health intervention research project, aimed to optimize interventions to be rolled out to participants to improve learning outcomes.

Functional Data Analysis for Gait Data 2017 – 2018

- Worked as a Research Assistant jointly with Department of Physiotherapy, University of Stellenbosch, Capetown and Department of Statistics, University of Calcutta.

- The project aimed to understand the effect of ITBS on gait of runners. The goal was to find out clinically relevant differences in the gait of runners affected by ITBS compared to the gait of unaffected runners.

e-Pathshala online lectures

2016 – 2017

- Presented lectures on Principal Component Analysis, Factor Analysis, Bayesian Analysis, and Multivariate Analysis as a part of University of Calcutta and Government of India's initiative to impart higher education for everyone.

Honors And Awards

Texas A & M University

2021 – Current

- **Third Place Award at 2026 Best of Statistical Science (BOSS) Workshop** for presenting *Nested Atoms Model with Application to Clustering Big Population-Scale Single-Cell Data* (April, 2026).
- **Emanuel Parzen Graduate Research Fellowship Award** for demonstrating exemplary research, above and beyond what is expected for graduation, awarded by the Department of Statistics, Texas A&M University (August, 2025).
- **Institute of Mathematical Statistics Hannan Graduate Student Travel Award** for presenting *Global-Local Dirichlet Processes for Clustering Grouped Data in the Presence of Group-Specific Idiosyncratic Variables* at the 14th International Conference on Bayesian Nonparametrics, UCLA, USA, 2025 (June 2025).
- **Bayesian Nonparametrics - International Society for Bayesian Analysis Travel Award** for presenting *Global-Local Dirichlet Processes for Clustering Grouped Data in the Presence of Group-Specific Idiosyncratic Variables* at the 14th International Conference on Bayesian Nonparametrics, UCLA, USA, 2025, awarded by National Science Foundation (June 2025).
- **2025 SETCASA Poster Competition Golden Award** for *Global-Local Dirichlet Processes for Clustering Grouped Data in the Presence of Group-Specific Idiosyncratic Variables*, awarded by the Department of Statistics, Texas A&M University (May, 2025).
- **Section on Bayesian Statistical Science Student Paper Award 2024** for presenting *Graphical Dirichlet Process for Clustering Non-Exchangeable Grouped Data*, awarded by the **American Statistical Association** (August 2024).
- **International Society for Bayesian Analysis Travel Award** for presenting *Graphical Dirichlet Process for Clustering Non-Exchangeable Grouped Data* at the ISBA World Meeting, Venice, Italy (July 2024).
- **Poster award, 6th Annual Symposium Cancer Research: Basic Science to Bioinformatics** for *Joint Bayesian estimation of cell dependence and gene associations in spatially resolved transcriptomic data* (November 2023).
- **International Society for Bayesian Analysis Travel Award** for presenting *Graphical Dirichlet Process for Clustering Non-Exchangeable Grouped Data* at Bayesian Nonparametrics Networking Workshop, Melbourne, Australia (September 2023).
- **William S. Connor Award** for the most outstanding among those students passing the preliminary examinations during the current year, awarded by the Department of Statistics, Texas A&M University (August 2022).

University of Calcutta

2015 – 2017

- Gold Medal in M.Sc in Statistics for being the University Topper.
- R.C Bose Book Prize for securing First position in semester I and semester II of M. Sc course (awarded by the Calcutta Statistical Association, 2016).
- DST INSPIRE Scholarship, awarded to top undergraduate students for pursuing graduate courses in Natural and Basic Sciences.

Presidency University

2012 – 2015

- Gold Medal in B.Sc Honours in Statistics for being the University Topper.

- Prof. Anil Kumar Bhattacharya Memorial Prize for being First in faculty of Science.
- Dr. Syama Prasad Mookherjee Gold Medal for Scholastic Record UG Major.
- DST INSPIRE Scholarship, awarded to top high school students for pursuing undergraduate courses in Natural and Basic Sciences.